Managed Agents vs Perplexity Computer

Which execution backend should Valor adopt?

April 2026

Governing Thought

They solve **different problems**.

MA is a programmable execution engine.

Perplexity Computer is an autonomous digital worker.

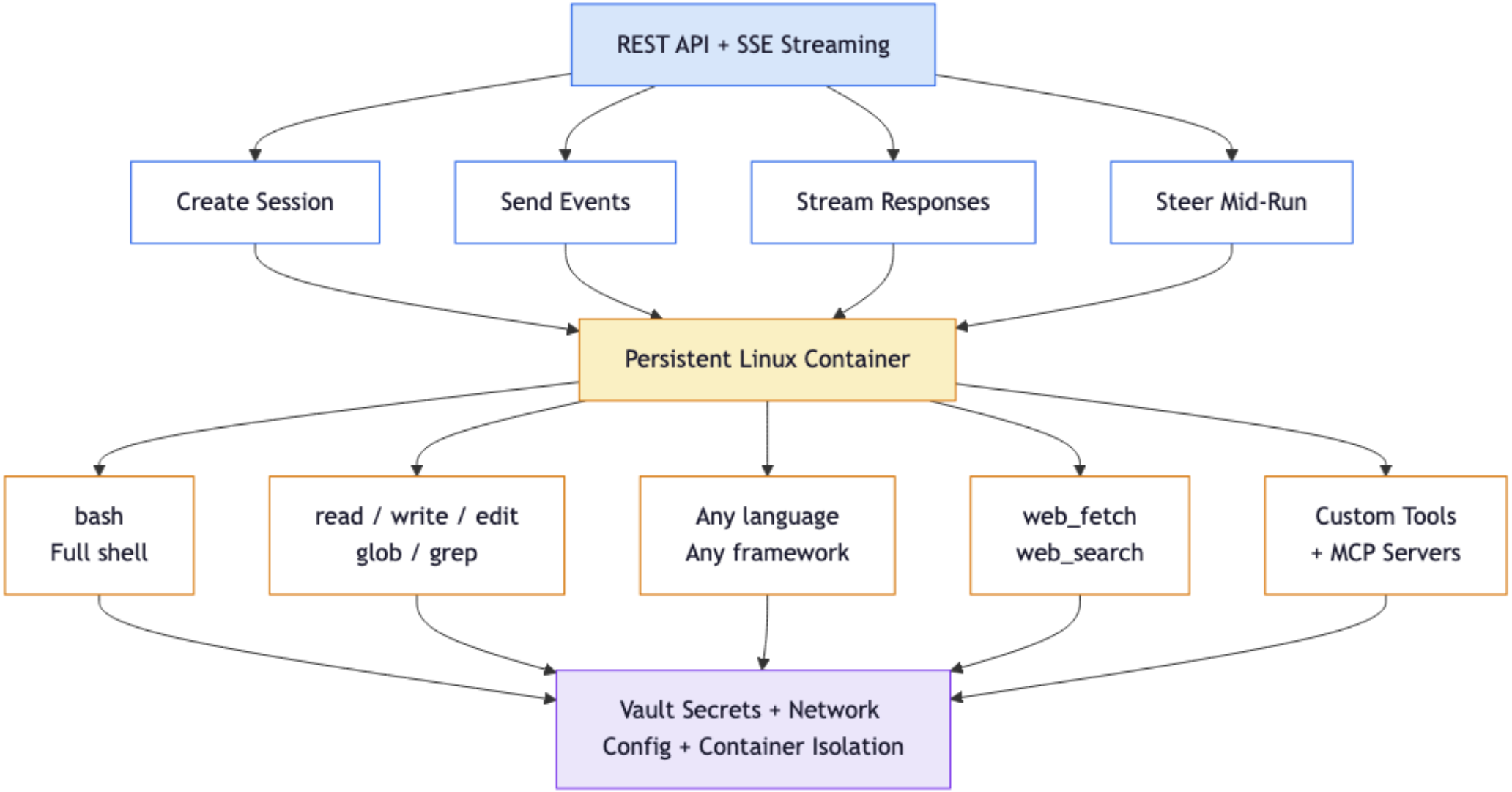
For Valor: **MA is the backend. Perplexity is not.**

What Are We Comparing?

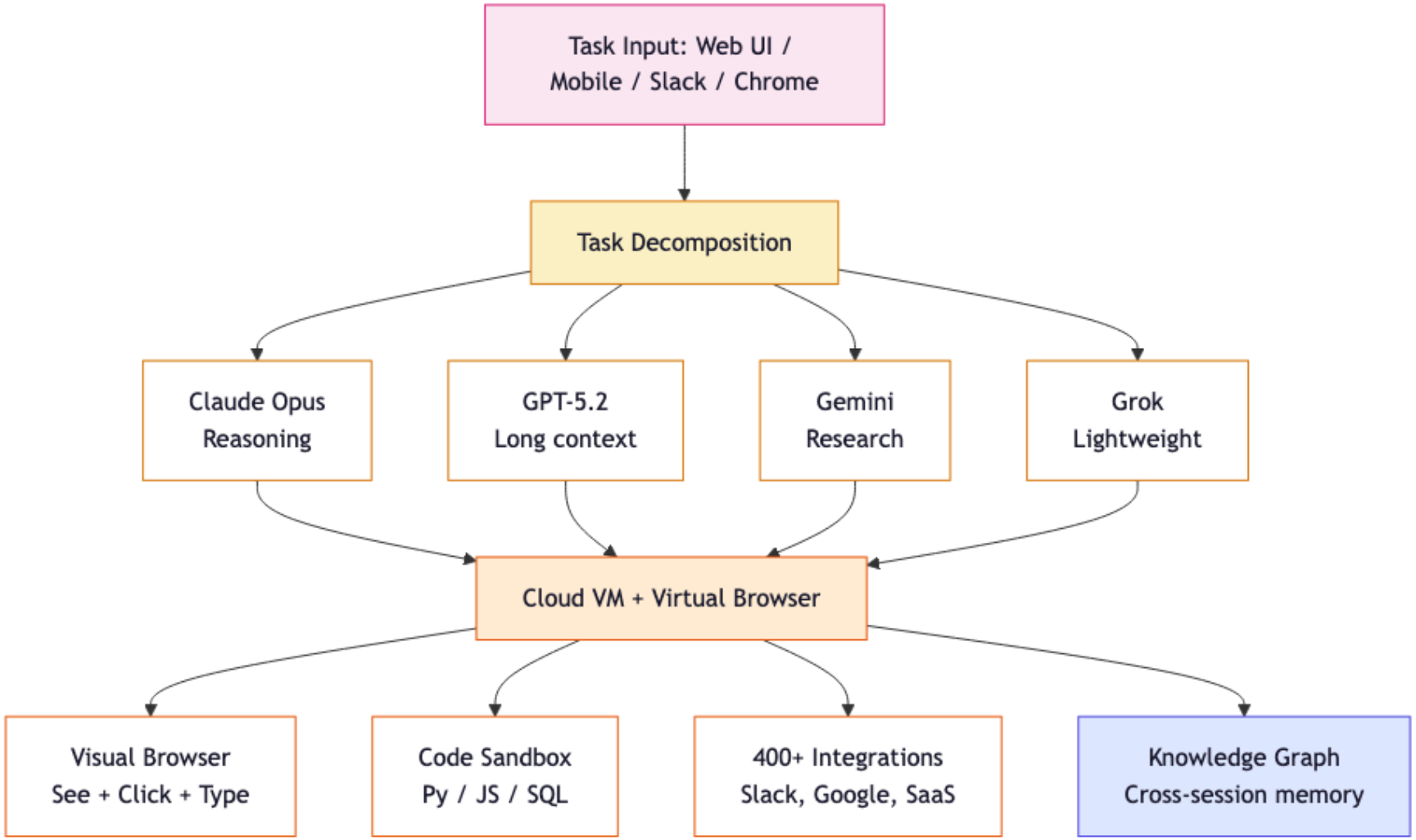
	CLAUDE MANAGED AGENTS	PERPLEXITY COMPUTER
Launched	April 2026 (beta)	February 2026
What it is	API-first containerized agent runtime	UI-first multi-model digital worker
Primary interface	REST API + SSE streaming	Web UI, mobile app, Slack
How it works	Runs code in Linux containers	Sees and clicks rendered web pages
Models	Claude family	19+ models (Claude, GPT, Gemini, Grok)
Pricing	Per-ms + per-token + per-search	$200/mo subscription + opaque credits
Target user	Developers building agent systems	Knowledge workers automating tasks

Two products that happen to both be called "AI agents" but share almost no overlap in architecture, audience, or use case.

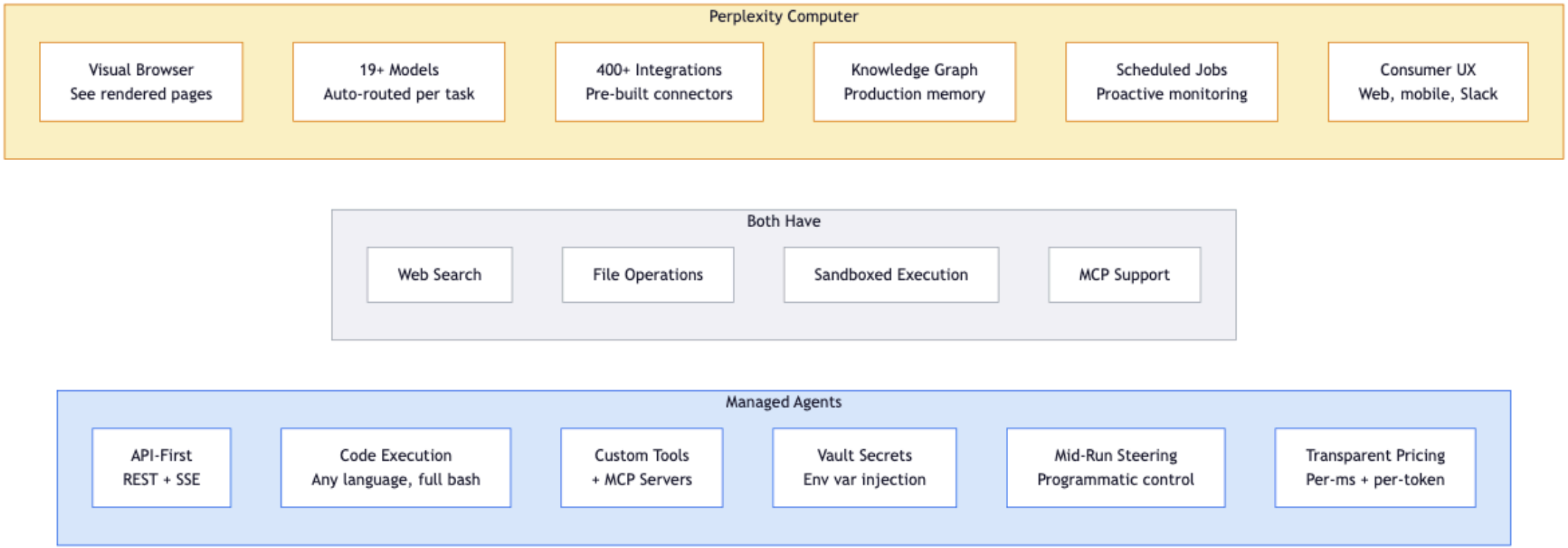
Managed Agents: The Developer's Backend



Perplexity Computer: The Digital Worker



Feature Comparison: What Each Brings



Common ground is narrow (web search, files, sandboxing, MCP). The unique capabilities point in opposite directions — MA toward **developer orchestration**, Perplexity toward **end-user automation**.

Architecture: A Fork, Not a Spectrum

ASPECT	MANAGED AGENTS	PERPLEXITY COMPUTER
Runtime	Linux container	Cloud VM + virtual browser
Agent sees	Text, files, structured data	Rendered pixels (vision-language model)
Agent acts	Shell commands, file writes	Clicks, types, scrolls, navigates
Isolation	Per-session container	Per-task container + browser
Persistence	Environments reused; sessions ephemeral	Knowledge graph persists across tasks
Networking	Configurable (restricted/unrestricted)	Unrestricted web, sandboxed from internal

MA is a **developer tool** -- it exposes a container. Perplexity Computer is a **digital worker** -- it exposes a virtual desktop.

The Orchestration Gap

CAPABILITY	MANAGED AGENTS	PERPLEXITY COMPUTER
Task creation	`POST /v1/sessions`	Web UI, mobile, Slack only
Streaming	SSE event stream	None (programmatic)
Mid-execution steering	`user.message` events	Not available
Custom tool routing	JSON schema, client-executed	No mechanism
Event history	Full retrieval via API	UI-only
Session lifecycle	create/stream/steer/archive	Submit task, get results

```
# MA: full programmatic control
session = client.beta.sessions.create(agent=agent.id, environment_id=env.id)
client.beta.sessions.events.send(session.id, events=[...])
with client.beta.sessions.events.stream(session.id) as stream:
    for event in stream: ...
```

Perplexity's **Agent API** is programmable but is a **separate product** -- it does not control

Tool Ecosystem

CATEGORY	MANAGED AGENTS	PERPLEXITY COMPUTER
Shell	Full bash	Not documented
File ops	read/write/edit/glob/grep	FUSE filesystem
Code execution	Any language (container)	Python, JS, SQL (sandbox)
Web search	Text results	Multi-model research + citations
Browser	None built-in (install Playwright)	Full visual browser
Custom tools	JSON schema, client-executed	Via separate Agent API
MCP servers	Native support	Local + remote
Integrations	None (bring your own)	400+ pre-built
Package mgmt	Pre-install in environment	Per-session install

MA goes **deep** on development. Perplexity goes **wide** on business workflows.

Web Browsing: Perplexity's One Advantage

CAPABILITY	MANAGED AGENTS	PERPLEXITY COMPUTER
Page rendering	Text extraction only	Full visual (VLM)
JS/SPA support	Via Playwright in container	Native
Form filling	Via Playwright	Native visual interaction
Screenshots	Via Playwright	Native
Setup required	Install Chromium in environment	None
Determinism	High (programmatic control)	Lower (visual interpretation)

For Valor, this gap matters little: BUILD/TEST need code tools, not browsers. Frontend testing uses Playwright in containers regardless. Research uses `web_search` + `web_fetch`.

Code Execution: MA's Clear Win

ASPECT	MANAGED AGENTS	PERPLEXITY COMPUTER
Languages	Any (container)	Python, JS, SQL
Shell access	Full bash	Not documented
Git	Full (push/pull/branch via vault tokens)	Struggles (21K credits to push to GitHub)
Test frameworks	Any (pytest, jest, playwright)	Not designed for this
Build tools	Any (npm, pip, cargo, make)	Limited
Environment	Pre-baked, reusable	Reinstalled per session

Perplexity Computer was built to automate business tasks, not write code. Using it for software development is like using a web browser to compile C++ -- technically possible, practically painful.

Pricing: Predictable vs Opaque

20 BUILD sessions/month (Valor's typical workload)

	MANAGED AGENTS	PERPLEXITY COMPUTER
Runtime	$3.20	N/A
Tokens	$60-160	Included in credits
Web searches	$1.00	Included
Monthly total	**$65-165**	**$200-500+**

MA billing is transparent: per-ms runtime + per-token + per-search.

Perplexity credits are opaque: no published per-task rates, complexity-based consumption. One documented case: a 40-minute codebase scan burned 21,000 credits (2x the monthly allotment).

Memory Integration

	MANAGED AGENTS	PERPLEXITY COMPUTER
Built-in memory	Research preview	Production knowledge graph
External memory	Custom tools query Redis, etc.	MCP connections
Stealth injection	Yes (via custom tool results)	No mechanism
Post-session extraction	Pull transcript via API	No programmatic access

For Valor's subconscious memory system, MA is the only option:

1. Custom tools (run_tests, git_commit) route through Valor
2. Valor queries Redis for relevant memories
3. Returns tool results + injected `<thought>` blocks
4. Agent never knows -- subconscious by design

Perplexity Computer's memory is internal and inaccessible to external orchestrators.

Valor Integration Assessment

MA: High viability (2-3 week PoC)

- Worker creates MA sessions instead of local processes
- Vault replaces local `.env` for secrets
- Custom tools enable stealth memory injection
- SSE enables real-time monitoring + steering
- Post-session transcript extraction for memory learning

Perplexity Computer: Not viable

- No API for task creation, streaming, or steering
- No custom tool injection
- No secret management
- No way to integrate Valor's memory system

Decision Matrix

CRITERION	WEIGHT	MA	PERPLEXITY COMPUTER
Programmatic orchestration	30%	10	2
Code execution	20%	10	5
Cost predictability	15%	9	4
Memory integration	15%	8	2
Web browsing	10%	5	10
Session management	10%	9	4
Weighted total	**100%**	**8.9**	**3.7**

The gap is not close. MA was designed for exactly what Valor needs. Perplexity Computer was designed for something else entirely.

What to Watch

SIGNAL	IMPACT	URGENCY
Perplexity ships Computer API	Would make Computer viable for orchestration	Watch
MA visual browser tool	Closes the web browsing gap entirely	Medium
MA memory GA	Simplifies memory architecture	High
MA hook-equivalent API	Makes subconscious memory fully portable	High
Perplexity credit transparency	Better cost comparison	Low

Trigger points

- **Perplexity Computer API**: re-evaluate for hybrid (MA for code, Computer for web research)
- **MA memory + hooks GA**: full migration to MA becomes viable
- **MA visual browser**: Perplexity Computer becomes irrelevant for Valor

Recommendation

Adopt MA as Valor's execution backend for BUILD and TEST.

Evaluate Perplexity Agent API as a supplementary research tool.

Do not adopt Perplexity Computer -- it solves a different problem.

The moat is orchestration and memory, not execution.